MSG No. 192.252 P_c6/mcc [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6/mm'm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,13,14,15,16,17,18,31,32,33,34,35,36,43,44,45,46,47,48]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[7,8,9,10,11,12,19,20,21,22,23,24,25,26,27,28,29,30,37,38,39,40,41,42]`

Table 2: Wyckoff site: `2b`, site symmetry: `6/m'm'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,3,5,14,16,18,34,35,36,43,44,45]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[2,4,6,13,15,17,31,32,33,46,47,48]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[7,8,9,22,23,24,26,28,30,37,39,41]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[10,11,12,19,20,21,25,27,29,38,40,42]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,3,5,10,11,12,38,40,42,43,44,45]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[2,4,6,7,8,9,37,39,41,46,47,48]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[13,15,17,22,23,24,26,28,30,31,32,33]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[14,16,18,19,20,21,25,27,29,34,35,36]`

Table 5: Wyckoff site: `4e`, site symmetry: `6m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6,43,44,45,46,47,48]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,9,10,11,12,37,38,39,40,41,42]`
3	$[0,\ 0,\ -z]$	`[13,14,15,16,17,18,31,32,33,34,35,36]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[19,20,21,22,23,24,25,26,27,28,29,30]`

Table 6: Wyckoff site: `6f`, site symmetry: `mm'm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,13,16,31,35,43,47]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,5,14,17,33,34,45,46]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,6,15,18,32,36,44,48]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[7,11,19,23,25,28,37,40]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[8,12,20,24,27,30,39,42]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,21,22,26,29,38,41]

Table 7: Wyckoff site: 6g, site symmetry: m'm'm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,4,7,11,37,40,43,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[2,5,9,10,38,41,45,46]
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[3,6,8,12,39,42,44,48]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[13,16,19,23,25,28,31,35]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[14,17,21,22,26,29,33,34]
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[15,18,20,24,27,30,32,36]

Table 8: Wyckoff site: 8h, site symmetry: 3m'.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5,43,44,45]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6,46,47,48]
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	[7,8,9,37,39,41]
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[10,11,12,38,40,42]
5	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[13,15,17,31,32,33]
6	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[14,16,18,34,35,36]
7	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	[19,20,21,25,27,29]
8	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[22,23,24,26,28,30]

Table 9: Wyckoff site: 12i, site symmetry: 2m'm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4,43,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5,45,46]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6,44,48]
4	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[7,11,37,40]
5	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[8,12,39,42]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[9,10,38,41]
7	$\left[\frac{1}{2}, 0, -z\right]$	[13,16,31,35]
8	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[14,17,33,34]
9	$\left[0, \frac{1}{2}, -z\right]$	[15,18,32,36]
10	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[19,23,25,28]
11	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[20,24,27,30]
12	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[21,22,26,29]

Table 10: Wyckoff site: `12j`, site symmetry: `m2'm'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,16,31,47]`
2	$[x,\ x,\ 0]$	`[2,17,34,45]`
3	$[0,\ x,\ 0]$	`[3,18,32,48]`
4	$[-x,\ 0,\ 0]$	`[4,13,35,43]`
5	$[-x,\ -x,\ 0]$	`[5,14,33,46]`
6	$[0,\ -x,\ 0]$	`[6,15,36,44]`
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[7,23,25,40]`
8	$\left[0,\ x,\ \frac{1}{2}\right]$	`[8,24,27,42]`
9	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[9,22,29,38]`
10	$\left[x,\ x,\ \frac{1}{2}\right]$	`[10,21,26,41]`
11	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[11,19,28,37]`
12	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[12,20,30,39]`

Table 11: Wyckoff site: `12k`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,7,40,47]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,10,41,45]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,8,42,48]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,11,37,43]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,9,38,46]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,12,39,44]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[13,19,28,35]`
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[14,22,29,33]`
9	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[15,20,30,36]`
10	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[16,23,25,31]`
11	$\left[x,\ x,\ \frac{3}{4}\right]$	`[17,21,26,34]`
12	$\left[0,\ x,\ \frac{3}{4}\right]$	`[18,24,27,32]`

Table 12: Wyckoff site: `12l`, site symmetry: `mm'2'`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,16,35,43]`
2	$[-x,\ x,\ 0]$	`[2,17,33,46]`
3	$[-2x,\ -x,\ 0]$	`[3,18,36,44]`
4	$[-x,\ -2x,\ 0]$	`[4,13,31,47]`
5	$[x,\ -x,\ 0]$	`[5,14,34,45]`
6	$[2x,\ x,\ 0]$	`[6,15,32,48]`
7	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[7,23,28,37]`
8	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[8,24,30,39]`
9	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,22,26,41]`
10	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[10,21,29,38]`
11	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[11,19,25,40]`
12	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[12,20,27,42]`

Table 13: Wyckoff site: `12m`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11,40,43]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,9,41,46]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,12,42,44]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,7,37,47]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,10,38,45]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,8,39,48]`
7	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[13,23,28,31]`
8	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[14,21,29,34]`
9	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[15,24,30,32]`
10	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[16,19,25,35]`
11	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[17,22,26,33]`
12	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[18,20,27,36]`

Table 14: Wyckoff site: `24n`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,47]`
2	$[x,\ x,\ z]$	`[2,45]`
3	$[0,\ x,\ z]$	`[3,48]`
4	$[-x,\ 0,\ z]$	`[4,43]`
5	$[-x,\ -x,\ z]$	`[5,46]`
6	$[0,\ -x,\ z]$	`[6,44]`
7	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[7,40]`
8	$\left[0,\ x,\ \frac{1}{2}-z\right]$	`[8,42]`
9	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	`[9,38]`
10	$\left[x,\ x,\ \frac{1}{2}-z\right]$	`[10,41]`
11	$\left[-x,\ 0,\ \frac{1}{2}-z\right]$	`[11,37]`
12	$\left[0,\ -x,\ \frac{1}{2}-z\right]$	`[12,39]`
13	$[-x,\ 0,\ -z]$	`[13,35]`
14	$[-x,\ -x,\ -z]$	`[14,33]`
15	$[0,\ -x,\ -z]$	`[15,36]`
16	$[x,\ 0,\ -z]$	`[16,31]`
17	$[x,\ x,\ -z]$	`[17,34]`
18	$[0,\ x,\ -z]$	`[18,32]`
19	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[19,28]`
20	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[20,30]`
21	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[21,26]`
22	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[22,29]`
23	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[23,25]`
24	$\left[0,\ x,\ z+\frac{1}{2}\right]$	`[24,27]`

Table 15: Wyckoff site: 24o, site symmetry: .m'.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,43]
2	$[-x,\ x,\ z]$	[2,46]
3	$[-2x,\ -x,\ z]$	[3,44]
4	$[-x,\ -2x,\ z]$	[4,47]
5	$[x,\ -x,\ z]$	[5,45]
6	$[2x,\ x,\ z]$	[6,48]
7	$\left[-x,\ -2x,\ \frac{1}{2}-z\right]$	[7,37]
8	$\left[2x,\ x,\ \frac{1}{2}-z\right]$	[8,39]
9	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	[9,41]
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[10,38]
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[11,40]
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[12,42]
13	$[-x,\ -2x,\ -z]$	[13,31]
14	$[x,\ -x,\ -z]$	[14,34]
15	$[2x,\ x,\ -z]$	[15,32]
16	$[x,\ 2x,\ -z]$	[16,35]
17	$[-x,\ x,\ -z]$	[17,33]
18	$[-2x,\ -x,\ -z]$	[18,36]
19	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	[19,25]
20	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	[20,27]
21	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	[21,29]
22	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[22,26]
23	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[23,28]
24	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[24,30]

Table 16: Wyckoff site: 24p, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,16]
2	$[x-y,\ x,\ 0]$	[2,17]
3	$[-y,\ x-y,\ 0]$	[3,18]
4	$[-x,\ -y,\ 0]$	[4,13]
5	$[-x+y,\ -x,\ 0]$	[5,14]
6	$[y,\ -x+y,\ 0]$	[6,15]
7	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	[7,23]
8	$\left[y,\ x,\ \frac{1}{2}\right]$	[8,24]
9	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	[9,22]
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	[10,21]
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	[11,19]
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[12,20]
13	$\left[x,\ y,\ \frac{1}{2}\right]$	[25,40]
14	$\left[x-y,\ x,\ \frac{1}{2}\right]$	[26,41]
15	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[27,42]
16	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[28,37]
17	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[29,38]

continued ...

Table 16

No.	position	mapping
18	$\left[y,\, -x+y,\, \frac{1}{2}\right]$	[30,39]
19	$[x-y,\, -y,\, 0]$	[31,47]
20	$[y,\, x,\, 0]$	[32,48]
21	$[-x,\, -x+y,\, 0]$	[33,46]
22	$[x,\, x-y,\, 0]$	[34,45]
23	$[-x+y,\, y,\, 0]$	[35,43]
24	$[-y,\, -x,\, 0]$	[36,44]

Table 17: Wyckoff site: 24q, site symmetry: m'..

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	[1,40]
2	$\left[x-y,\, x,\, \frac{1}{4}\right]$	[2,41]
3	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	[3,42]
4	$\left[-x,\, -y,\, \frac{1}{4}\right]$	[4,37]
5	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	[5,38]
6	$\left[y,\, -x+y,\, \frac{1}{4}\right]$	[6,39]
7	$\left[x-y,\, -y,\, \frac{1}{4}\right]$	[7,47]
8	$\left[y,\, x,\, \frac{1}{4}\right]$	[8,48]
9	$\left[-x,\, -x+y,\, \frac{1}{4}\right]$	[9,46]
10	$\left[x,\, x-y,\, \frac{1}{4}\right]$	[10,45]
11	$\left[-x+y,\, y,\, \frac{1}{4}\right]$	[11,43]
12	$\left[-y,\, -x,\, \frac{1}{4}\right]$	[12,44]
13	$\left[-x,\, -y,\, \frac{3}{4}\right]$	[13,28]
14	$\left[-x+y,\, -x,\, \frac{3}{4}\right]$	[14,29]
15	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	[15,30]
16	$\left[x,\, y,\, \frac{3}{4}\right]$	[16,25]
17	$\left[x-y,\, x,\, \frac{3}{4}\right]$	[17,26]
18	$\left[-y,\, x-y,\, \frac{3}{4}\right]$	[18,27]
19	$\left[-x+y,\, y,\, \frac{3}{4}\right]$	[19,35]
20	$\left[-y,\, -x,\, \frac{3}{4}\right]$	[20,36]
21	$\left[x,\, x-y,\, \frac{3}{4}\right]$	[21,34]
22	$\left[-x,\, -x+y,\, \frac{3}{4}\right]$	[22,33]
23	$\left[x-y,\, -y,\, \frac{3}{4}\right]$	[23,31]
24	$\left[y,\, x,\, \frac{3}{4}\right]$	[24,32]

Table 18: Wyckoff site: 48r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x-y,\, x,\, z]$	[2]
3	$[-y,\, x-y,\, z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x+y,\, -x,\, z]$	[5]

continued ...

Table 18

No.	position	mapping
6	$[y,\, -x+y,\, z]$	[6]
7	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[7]
8	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[8]
9	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[9]
10	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[-x+y,\, -x,\, -z]$	[14]
15	$[y,\, -x+y,\, -z]$	[15]
16	$[x,\, y,\, -z]$	[16]
17	$[x-y,\, x,\, -z]$	[17]
18	$[-y,\, x-y,\, -z]$	[18]
19	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[19]
20	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[20]
21	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]
25	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[25]
26	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	[26]
27	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[27]
28	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[29]
30	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	[30]
31	$[x-y,\, -y,\, -z]$	[31]
32	$[y,\, x,\, -z]$	[32]
33	$[-x,\, -x+y,\, -z]$	[33]
34	$[x,\, x-y,\, -z]$	[34]
35	$[-x+y,\, y,\, -z]$	[35]
36	$[-y,\, -x,\, -z]$	[36]
37	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[37]
38	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	[38]
39	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[39]
40	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[40]
41	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[41]
42	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	[42]
43	$[-x+y,\, y,\, z]$	[43]
44	$[-y,\, -x,\, z]$	[44]
45	$[x,\, x-y,\, z]$	[45]
46	$[-x,\, -x+y,\, z]$	[46]
47	$[x-y,\, -y,\, z]$	[47]
48	$[y,\, x,\, z]$	[48]